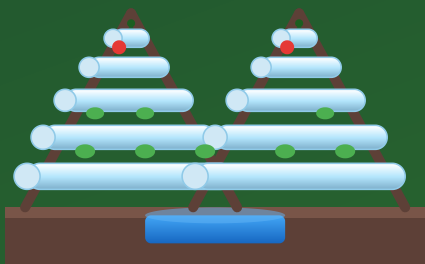


Do's & Don'ts

A practical field guide for running a healthy NFT A-Frame hydroponic system



Your NFT A-Frame System

2 A-Frames · 4 independent sides · 5 pipes per side · 6m per pipe.

Each side has its own pump and tank, allowing different crops and nutrient recipes on each side.

🌿 Lettuce

🌿 Mustard Spinach

🍓 Strawberry

30min

ROOTS DIE WITHOUT FLOW

24/7

PUMP MUST RUN

24°C

MAX WATER TEMP

1×

CHANGE AT A TIME

4–6wk

TANK CLEAN CYCLE

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The 3 Rules You Cannot Break

These rules protect your entire system. Breaking any one of them can kill your crop within hours.



CRITICAL — RULE 1 OF 3

The pump runs 24 hours a day, 7 days a week — no exceptions

In an NFT system, plant roots are always exposed to air inside the pipe. The only moisture they receive is from the thin film of water the pump pushes through. The moment the pump stops, that film disappears. Roots begin to dry out within minutes and die within 30 minutes — there is no recovery once roots are dead. Always keep a spare pump in stock. When a pump fails, you have 30 minutes to replace it.



Without water flow: Roots dry out and die in under 30 minutes. Dead roots cannot be revived. The entire plant must be replaced.



CRITICAL — RULE 2 OF 3

Never adjust EC and pH at the same time — one change at a time

EC (nutrient strength) and pH (acidity) are both important, but they must always be corrected one at a time. If you adjust both simultaneously, you cannot know which change caused which effect — and you risk overcorrecting in both directions, creating a solution that is more harmful than the original problem. Always fix pH first, wait 30 minutes, measure again, then adjust EC if still needed.



Dual correction risk: Overcorrection in both directions simultaneously often stresses plants far more than the original problem. Patience and one-step adjustments are essential.



CRITICAL — RULE 3 OF 3

Never let the reservoir tank run dry

When the tank runs dry, the pump has no water to move — it runs on air, overheats rapidly and burns out. This creates a double failure: no pump AND no water. Tank levels drop faster on hot days due to evaporation and when plants are feeding heavily. Check all 4 tanks every morning without fail. If a tank is low, top it up with a premixed nutrient solution — not plain water only.



Empty tank cascade failure: Dry pump → burned motor → no flow → dead roots. Two simultaneous failures that both need replacing.



What You Should Always Do

These 6 habits keep your system healthy, your water clean and your crops growing well.



SETUP

Keep a 1–2% downward slope on all pipes

Pipes must slope slightly downward from the inlet to the drain. This lets gravity pull the nutrient film along the pipe. Without a slope, water pools in the pipe, fully submerging roots and cutting off their oxygen — causing root rot.



Use a spirit level during installation. Check slope again after any pipe is moved or bumped.



SETUP

Use white or light-coloured pipes only

Dark-coloured pipes absorb heat from sunlight and raise the water temperature inside. Water above 24°C holds less dissolved oxygen, which means roots suffocate even when the pump is running. Light pipes also reduce internal algae growth.



Target water temperature: 18–22°C. Check it every morning with your EC meter if it has a temperature sensor.



ROUTINE

Always keep the reservoir tank covered

An uncovered tank allows sunlight in, which triggers algae blooms. Algae consumes the nutrients meant for your plants, clogs the pump filter, and produces substances harmful to roots. A lid also reduces evaporation and prevents debris from falling in.



Even a small gap lets in enough light for algae. Seal all openings around pipes entering the tank lid.



IMPORTANT

Measure your tap water EC before mixing

Tap water already contains dissolved minerals and will show an EC reading before any nutrients are added — often 0.2–0.5 mS/cm. Your starting point is not zero. If you ignore this, your final EC will be higher than intended and you risk salt stress in sensitive crops like strawberries.



Note tap water EC and subtract it when calculating how much nutrient to add.



ROUTINE

Flush and clean tanks every 4–6 weeks

Over time, salt build-up forms on tank walls and inside pipes, distorting EC readings and blocking flow. Biofilm on surfaces harbours root pathogens. Drain the tank, scrub with diluted white vinegar, rinse thoroughly with clean water, and refill with fresh nutrient solution.



Always clean between crop cycles. Do not plant a new crop into an uncleaned system from the previous one.



IMPORTANT

Record EC and pH readings every morning

A single reading tells you the state right now. A written log tells you the trend — whether your system is stable, drifting, or getting worse. Trends let you catch problems before plants show any visible symptoms, saving crops before damage occurs.



Use the monitoring log in this handleiding. Note date, EC, pH, water temp and any observations per tank.



What You Must Never Do

These 6 mistakes are the most common causes of crop loss, pump failure and root disease in NFT systems.



CRITICAL

Never fully submerge the roots in water

NFT stands for Nutrient Film Technique — the water is meant to be a thin film, not a pool. Roots must have open air above the film. Fully submerged roots have no access to oxygen and will develop root rot within 2–3 days, even if the water is clean.

- 💡 If water is pooling in the pipe, check for blocked drains or insufficient pipe slope.



CRITICAL

Never let roots grow into and block the drain

Healthy roots grow vigorously and can fill the entire pipe, reaching the drain opening and blocking it. Backed-up water floods the pipe and drowns all roots in that pipe. Check drain openings weekly, especially in the 3rd and 4th week of a crop cycle when roots are largest.

- 🔍 Look into the drain end of each pipe weekly. Trim roots that are approaching the opening.



IMPORTANT

Never mix Part A and Part B nutrients directly together

Most hydroponic nutrient systems come in two parts (A and B) that must be added separately to the tank. Mixing them together in concentrated form causes a chemical reaction that forms a white precipitate — the nutrients lock together and become permanently unavailable to plants.

- ✅ Add Part A to the tank water first, stir well, then add Part B. Never pour them together undiluted.



IMPORTANT

Never add large doses of pH Up or pH Down at once

pH adjusters are very concentrated. A single large dose can swing the pH 2–3 units past the target, overcorrecting into the opposite extreme. Add a small amount — about 1ml per 20 litres — stir, wait 10–15 minutes, then measure again before adding more.

- 🕒 pH takes time to stabilise in solution. Always wait 10–15 minutes after each addition before measuring again.



SETUP

Don't grow heavy or large-rooting crops without proper setup

Crops like tomatoes, cucumbers and peppers produce very large root masses that block water flow inside the pipe. Their heavy fruit also puts physical strain on the pipe supports. Standard NFT pipes (50mm) are not sized for these crops. They need wider pipes, stronger frames and individual support strings.

- ✅ Standard NFT is ideal for lettuce, spinach, herbs and strawberries. These are your best options for this system.



ROUTINE

Never plant a new crop into an uncleaned system

Old root material left in pipes after harvest decomposes and feeds Pythium — the water mould that causes root rot. Pythium spores spread rapidly through a shared water system. After every harvest, remove all root matter, flush pipes with clean water and disinfect with diluted 3% hydrogen peroxide before replanting.

- 💧 Disinfect: fill pipes with 3% H₂O₂ solution, leave for 30 min, then flush with fresh water before planting.



Warning Signs — Act Immediately

If you see any of these, stop what you are doing and investigate right away.



WHAT YOU SEE

Brown, slimy or foul-smelling roots

Cause: Pythium root rot. Drain the affected tank immediately. Remove all plants and root material. Scrub and disinfect the tank and pipes with diluted hydrogen peroxide. Check water temperature — Pythium thrives above 22°C. Only replant after full disinfection.



WHAT YOU SEE

EC is high but plants are yellow or weak

Cause: pH is wrong, not EC. When pH is out of range, plants cannot absorb nutrients even if EC is perfect — this is called nutrient lockout. Always check and fix pH first. Once pH is correct, re-measure EC and adjust if still needed.



WHAT YOU SEE

EC dropping faster than expected

Cause: Heavy feeding or rapid evaporation. First check the tank water level — a low level concentrates or dilutes readings. If level is fine, plants are feeding heavily. Top up with premixed nutrient solution, not plain water. Plain water lowers EC but does not replace nutrients.



WHAT YOU SEE

Green coating on pipes, tank walls or inside pipes

Cause: Algae bloom. Light is entering the system somewhere. Find and cover all gaps in pipe caps, tank lids and pipe openings. Clean affected surfaces with diluted bleach solution (1:100), then rinse thoroughly with clean water before reconnecting.



Morning Checklist — Do This Every Day

Six tasks that take less than 10 minutes and protect the entire system.

✓	TASK	WHAT TO CHECK	IF SOMETHING IS WRONG	LEVEL
☐	⚙️ All 4 pumps running	Steady flow, no noise, no leaks	Replace pump — you have 30 minutes	Critical
☐	💧 Tank water levels	Enough water, pump not running dry	Top up with premixed nutrient solution	Critical
☐	⚡ EC per tank	Within optimal range for that crop	Add nutrients (low) or dilute with water (high)	Important
☐	🧪 pH per tank	Within optimal range for that crop	Add pH Up or Down — small doses, wait 15 min	Important
☐	🌱 Visual plant check	Green leaves, upright, no wilting	Yellow = pH/nutrient issue. Wilting = flow issue	Important
☐	🔍 Check pipe	Water flowing freely, no root	Clear blockage, trim roots near drain	Routine